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Characterisation of structural performance and sustainability of hot mix asphalt mixtures containing RAP using Machine Learning

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ABSTRACT

This paper describes a new computational framework for characterising the effects of reclaimed asphalt pavement (RAP) parameters, sources and contents, on the structural performance of hot mix asphalt, in terms of dynamic modulus (E^*), of using machine learning (ML) techniques. The ML models were developed for five RAP content percentages (0%, 10%, 20%, 30%, and 40%) for three different sources. The ML models showed promising results with Taylor diagrams and R^2 values of 87% to 99% for all RAP sources and content levels. The results of pavement design using ML data showed the thicknesses of the asphalt mixture are slightly different depending on the RAP source. The same trend was found in the mixing and construction temperatures. Furthermore, analysis of sustainability showed that higher and stiffer RAP (higher E^*) results in a more sustainable asphalt mixture because of lower thicknesses, and consequently lower aggregate requirements and CO₂ emissions, for all ML methods and fuel types. Lastly, the incorporation of higher RAP content from any sources, the use of ML methods, and the adoption of cleaner fuel types results in massive material savings, higher energy efficiency, and improved sustainability in asphalt mix production for all RAP sources.

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Reclaimed asphalt pavement; asphalt pavement; machine learning; pavement design; sustainable asphalt mix; energy efficient asphalt mix

1. Introduction

Reclaimed asphalt pavement (RAP) has become one of the main sources of raw materials in the pavement industry. The promising results of the incorporation of RAP on the engineering properties and sustainability of asphalt mixtures can be found elsewhere (De Siqueira et al. 2025; Jeong, Kim, and Shen 2021; Taziani et al. 2017; Tsakoumaki and Plati 2024; Wang, Dong, and Zhang 2024). In addition, laboratory studies show that the incorporation of RAP may change the rheological characteristics of the resultant asphalt binder blends, which depends on RAP source, content, the amount of binder in the RAP, the degree of ageing, moisture content, the original binder type (i.e. asphalt binder performance grade [PG]), the volumetric properties of the asphalt mixtures, the asphalt mix/binder additive type and content, the wax content in the asphalt binder, and the petrochemical characteristics of crude oil (Fried et al. 2024; Jamshidi et al. 2015; Li et al. 2024). However, it is cumbersome to analyse the effects of all the variables mentioned above in the analysis of the rheological characteristics of asphalt binder blends and the engineering properties of asphalt mixtures containing RAP. Therefore, it is necessary that the asphalt industry benefits from new techniques and approaches, for example, machine learning (ML), in the modelling and analysis of RAP materials. For example, Botella et al. (2022)

proposed a computational framework to analyse the degree of binder activity of RAP as a percentage of indirect tensile strength test (ITS) using ML approaches, including sixth-degree multivariate polynomial regression with regularisation, and artificial neural network (ANN) and random forest regression. The ML models showed similar levels of precision; however, the ANN required less computational time than the other methods. In another study, Afshin and Behnood (2024) developed an ML-based framework to predict the moisture sensitivity of hot mix asphalt (HMA) containing RAP. Comparing the actual and predicted results of tensile stress ratios, it was concluded that ML models can be used as an accurate tool to analyse the risk of moisture susceptibility. In another study, Rondinella et al. (2023) showed that an ML technique (CatBoost) is a reliable approach to predict the ITS of HMA and cold mix asphalt containing RAP. In a detailed study, H. L. Nguyen and Quan Tran (2023) analysed the potential of ML techniques to predict laboratory rutting using a Hamburg wheel tracking machine based on different variables. A wide variety of ML methods were used, Analysis of the robustness of the models showed that extreme gradient boosting had the highest performance, and that temperature is the most critical factor for rutting. Although there are a lot of studies on RAP materials using different ML

approaches (Ayazi, Singh, and Kumar 2024; L. N. Nguyen et al. 2023), there is a gap in the knowledge regarding the characterisation of the effect of RAP sources and content on the structural performance of HMA using ML techniques. Additionally, the effect of ML techniques on the structural design of asphalt pavements containing RAP requires further investigation. This paper aims to fill these gaps.

2. Materials and methods

2.1. Asphalt Binder

The base asphalt binder, namely PG 64 was used as the control binder. The rheological characteristics of the base binder subjected under ageing conditions are shown in Table 1.

2.2. Aggregate

The aggregate type used for the mix production was granite with a specific heat capacity of 840 J/kg°C. The filler material was cement, and the selected aggregate

gradation is AC 14, which is the typical aggregate grading adopted for the wearing course nationally. The aggregate gradation is shown in Figure 1.

2.3. RAP

The RAP materials were attained from three different sources in Malaysia. The properties of the RAP binder from different sources are shown in Table 2. Prior to preparing the asphalt mixture, the RAP materials were pre-conditioned for 4 h at mixing temperatures.

2.4. Asphalt mixture

The fresh aggregate and different percentages of RAP were blended. The optimum binder content was 5% for all the mixture samples. After mixing, the loose mixed samples were poured into pans that were heated to compaction temperature. The pans were placed in the oven for 2 h to compaction temperature. Then, the samples were compacted using a slab compactor. The samples were then put at room temperature to cool down. Once cooled, the samples were cut into cylinder samples for testing. For each RAP content, three samples were produced. Table 3 shows the RAP content of each source. Three replicate specimens were prepared for each mix prepared from each RAP source and content.

2.5. Dynamic modulus test

A simple performance tester was used to measure the E^* of the asphalt mixtures according to AASHTO:TP62-03 (Figure 2). The test was typically conducted at six frequencies (25, 10, 5, 1, 0.5, and 0.1 Hz) and three temperatures (20, 40, and 50°C). The sample size was 150 mm in height and 100 mm in diameter. The average values of three replicates of each mix are considered as representative of the E^* of each mix.

2.6. Modelling of E^*

The models developed are based on the test parameters of the SPT setup and the RAP sources and content (Equation 1). In other words, the inputs of the models are the test temperatures, frequencies, and RAP content from different sources. The output of the models is the E^* of each RAP source. Obviously, the synergistic effects of the RAP source and content are reflected in the values of E^* .

Table 1. Rheological properties of PG 64.

Test properties	Results
Viscosity at 135°C (Pa.s)	0.47
$G^*/\sin \delta$ (kpa)	1.23
Failure temperature (°C)	66.40
Penetration test at 25°C (dmm)	88
Softening point (°C)	46

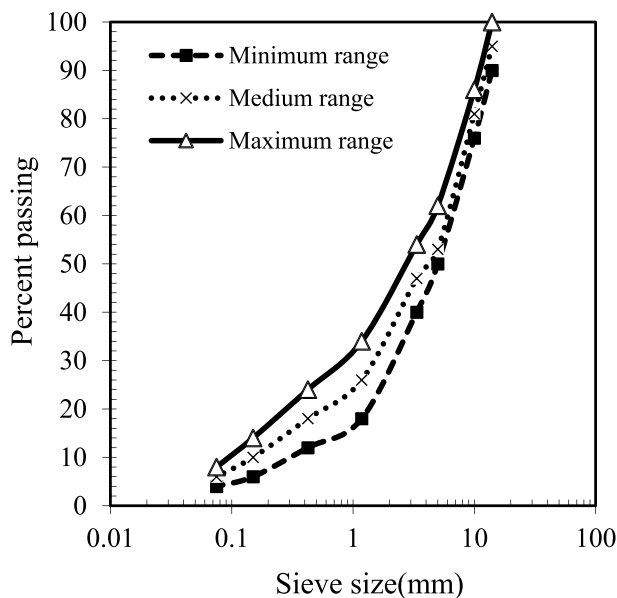


Figure 1. Aggregate gradation for asphalt mixture type.

Table 2. Rheological characteristics of RAP binders.

RAP source	Abbreviation	Penetration test at 25°C (dmm)	Softening point (°C)	Viscosity at 135°C (Pa.s)
North-South Expressway	NSE	14	67.5	3.06
Damansara Puchong Expressway	DPE	13	67.5	4.33
Public Work Department	PWD	11	72	2.14

Table 3. RAP content from the sources for the asphalt mixture production.

RAP source	RAP content (%)
NSE	10
	20
	30
	40
DPE	10
	20
	30
	40
PWD	10
	20
	30
	40


Figure 2. SPT set up and samples for testing.

$$E^* = f(T, fr, C) \begin{cases} PWD \\ DPE \\ NSE \end{cases} \quad (1)$$

where T is temperature, fr is frequency, and C is the RAP content.

There are different techniques for modelling E^* . In this study, ML is adopted to develop the functions of E^* . Therefore, the created current dataset includes four input parameters (temperature, frequency, RAP source, and content), with the output value or target being E^* . The initial preprocessing procedure involves encoding the qualitative parameter (RAP source) into a quantitative one. The dataset contains 270 data (observer) samples, including 3 temperature $\times$ 6 frequency $\times$ 3 RAP source $\times$ 5 RAP content sets, which are then split into training and testing sections with 231 sets for training and 39 sets for testing. To gain better insights into the effects of different RAP sources, the output results are also separated based on RAP sources, as shown in Equation (1). The details of the ML techniques adopted are explained in the following section.

2.7. ML techniques

2.7.1. Gaussian process regression

ML, which is a subfield of artificial intelligence (AI), employs data-driven modelling approaches to uncover patterns within datasets. The primary objective of regression processes is to minimise the error, defined

as the difference between predicted and actual outputs, which is the main focus in supervised learning approaches (Nasrin, Pourkamali-Anaraki, and Peterson 2024). Among the various modelling techniques, the Gaussian process (GP) model is versatile and is often employed for both continuous and discrete data. Specifically, the Gaussian process regression (GPR) model is adapted to handle continuous data for regression purposes. In GPR, the underlying assumption is that the data-generating process follows a Gaussian distribution over all possible functions. This assumption allows for an accurate measurement of uncertainty associated with predictions (Schulz, Speekenbrink, and Krause 2018).

The advantages of the GPR model lie in its flexibility and inherent capability to measure uncertainty in predictions under various conditions. The relationship between input parameters (predictors) and output parameters (responses) is illustrated in Equation 2, where a Gaussian distribution with a mean value of zero and a variance of σ_n^2 is used (El-Hakim et al. 2024).

$$y = f(x) \sim N(0, K(x, x_n), \sigma_n^2 I_n) \quad (2)$$

While $K(x, x')$ is the kernel function that ensures the main function is smoother, a significant advantage of using the Gaussian prior assumption is that functions can be conveniently specified by a mean function $f(x)$ and a covariance (or kernel) function. The kernel function typically contains several free parameters, known as hyperparameters, which define the prior distribution on $f(x)$ (Taki et al. 2018).

2.7.2. Artificial neural network

The artificial neural network (ANN) enhanced in the current study is based on a multilayer perceptron feed-forward layer network (MLP-FF). This network includes one input layer, one hidden layer with several predefined neurons, and one output layer, which is shown in Figure 3 (Razi and Athappilly 2005). The ANN has become a powerful tool for modelling complex relationships across various domains, including cleaner production. ANN models, especially multilayer perceptron (MLP) and radial basis function (RBF) networks, demonstrate superior performance in predicting outputs by learning from large datasets. In the context of cleaner production, an ANN can model the relationship between input variables (e.g. production techniques, resource usage) and output variables (e.g. environmental impact, economic performance). The general formulation of an ANN for regression can be expressed as in Equation 3:

$$y = f\left(\sum x_i w_i + b\right) \quad (3)$$

where y is the predicted output, x_i are the input variables, w_i are the weights, b is the bias, and f is the activation function.

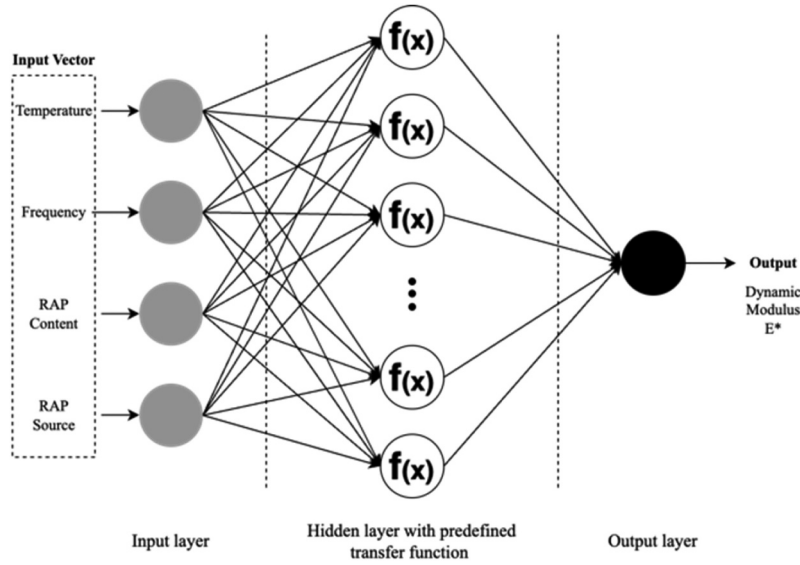


Figure 3. Illustration of the ANN architecture implemented in the current study.

The network is trained using algorithms such as backpropagation to minimise the error between predicted and actual outputs, often measured by metrics such as mean squared error (MSE). Backpropagation, short for ‘backward propagation of errors’, is a supervised learning algorithm that calculates the gradient of the loss function with respect to the network’s weights (Premalatha and Valan Arasu 2016). By iteratively adjusting the weights, backpropagation reduces the error and improves the model’s accuracy. Recent studies have shown the effectiveness of ANNs in optimising cleaner production processes, leading to significant improvements in both environmental and economic performance.

2.7.3. Kernel

The method of Gaussian kernel regression, referred to as the kernel in this study, is a form of nonlinear regression enhanced by random feature expansion approximation to evaluate predictions on large datasets (Le, Sarlós, and Smola 2013). Random feature expansion is particularly practical for big data applications with large training sets but can also be applied to smaller datasets that fit in memory. After mapping the predictor data into a high-dimensional space, the kernel regression algorithm searches for an optimal function that deviates from each response data point by values no greater than the epsilon margin.

Some regression problems cannot be described adequately using a linear model. In such cases, we obtain a nonlinear regression model using a nonlinear kernel function (Equation 4):

$$K(x, y) \approx T(x)T(y) \quad (4)$$

where $T(x)$ maps x in $\mathbb{R}^p$ to a high-dimensional space $\mathbb{R}^m$. The random kitchen sinks (Rahimi and Recht

2007) scheme uses the random transformation (Equation 5):

$$T(x) = m^{-\frac{1}{2}} \exp(iZx) \quad (5)$$

where $Z \in \mathbb{R}^{m \times p}$ is a sample drawn from $N(0, \sigma^{-2})$, and σ is a kernel scale.

2.7.4. Decision tree

Decision tree (DT) algorithms are among the most widely implemented ML techniques (Golafshani et al. 2024; Sarker 2021). They are well-known non-parametric supervised methods suitable for both regression and classification tasks. The algorithm starts at the root node and sorts down the tree to the leaf nodes, as illustrated in Figure 4. The process of growing branches corresponding to attribute values continues until a stopping criterion is met, forming leaf nodes that hold predictions for the target output values. In the following study, the DT method will be referred to as ‘tree’. The most popular criterion for impurity detection is the ‘gini’ index, which can be enhanced by the entropy equation (Pedregosa et al. 2012) to reduce overfitting and improve the model’s generalisation performance.

2.8. Accuracy metrics

ML algorithms are inherently complex, and datasets with multiple predictors can make the system’s complexity immeasurable. The best-performing ML model is the one that best describes the phenomenon that shows the same trend(s) or results under practical conditions. Hence, the adoption of proper assessment metric(s) is essential (Naser and Alavi 2023). In this regard, over 30 metrics have been introduced, as listed by Naser and Alavi (2023). However, multi-criteria

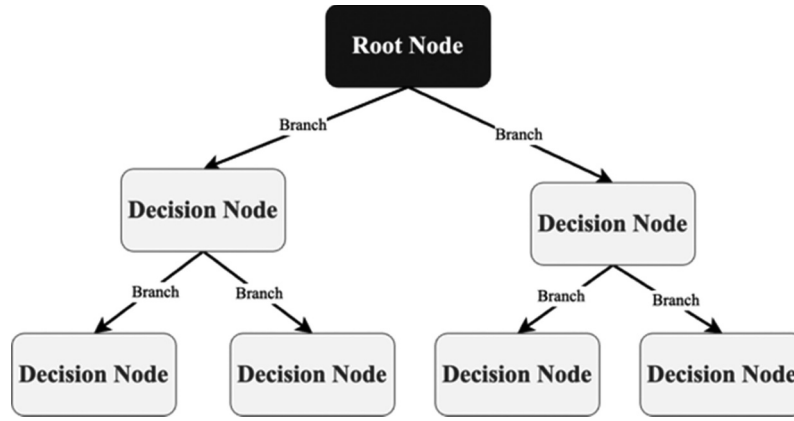


Figure 4. The schematic of the decision tree.

verification that incorporates several metrics is also recommended (Narimani et al. 2024).

A widely used statistic criterion for assessing models is the root mean squared error (RMSE) (Equation 6), especially for normal (Gaussian) errors (Hodson 2022). Taking the root does not affect the relative ranks of the models, but it produces a metric with the same units as the input data.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (Y_{\text{pre}} - Y_{\text{act}})^2} \quad (6)$$

For nonlinear regression models, the coefficient of determination (R^2) is another popular metric. However, it can become highly complicated when it produces a negative value instead of a value between 0 and 1. This issue arises due to the dependency of R^2 in nonlinear regression, as noted by some scholars (Nagelkerke 1991). Kvålseth (1985) provided comprehensive analysis and identified up to eight formulas available for determining R^2 (Equation 7). For the current study, the popular Equation 7 will be used, as it is considered promising.

$$R^2 = 1 - \frac{\sum_{i=1}^n (Y_{\text{pre}} - \bar{Y}_{\text{pre}})^2}{\sum_{i=1}^n (Y_{\text{act}} - \bar{Y}_{\text{act}})^2} \quad (7)$$

In Equations 6 and 7, the values denoted by Y_{act} , Y_{pre} , $\bar{Y}_{\text{act}}$, and $\bar{Y}_{\text{pre}}$ are the actual and expected values and actual and predicted mean values, respectively.

2.9. Taylor Diagram

The Taylor diagram (TD), introduced by Taylor (2001), offers a concise way to assess multiple model performances in one diagram. It combines three accuracy metrics, using the law of cosines to show the relationship between the standard deviations, correlation, and centred root mean squared error (CRMSE)

of the model outputs and reference data. Plotted in a polar coordinate system, the TD simplifies comparing various model outputs by representing them as distinct points, with their distance from the origin representing the CRMSE (Izzaddin et al. 2024).

2.10. Sensitivity analysis

The previous sensitivity analysis did not clearly define the effects of all input variables. To measure the significance of all predictors in one diagram (Ehsani et al. 2023; Mrzygłód et al. 2020) introduced a novel method that involves calculating the ratio of the model's error when excluding a specific (i-th) variable to the error when all variables are included. This ratio effectively demonstrates the importance of the i-th variable in the model's predictive accuracy, indicating which parameters the model is most sensitive to (Al-Jarazi et al. 2024). Equation 8 is applicable to the present study to determine the variable importance ratio (VIR), denoted as W_i :

$$W_i = \frac{\text{MSE}_i}{\text{MSE}} \quad (8)$$

where W_i is the VIR measured of the i-th variable, MSE is the mean squared error produced from the model when all predictors are included, and MSE_i is the mean squared error produced when the i-th variable is excluded from the model.

2.11. Pavement design

Based on the outputs of the model, an asphalt pavement with and without mechanical stabilisation was designed for a 6×10^6 equivalent single-axle load. The design subgrade California bearing ratio was 6% for the pavement design. The base layer consisted of high-quality gravel with an elastic modulus of 350 MPa. The pavement design was carried out based on AUSTROADS Guide 2 (Moffatt 2017).

2.12. Analysis of sustainability

Analysis of sustainability was carried out based on energy requirements and carbon footprint emissions for a dual carriageway 10-km length of road. Each carriageway has one lane (3.65 m). The volume of asphalt mix is calculated by multiplying the length of the road, the width of the two-lane road, and the thickness specified in the pavement design. The mass of the asphalt mix is then determined by multiplying the calculated volume by the material's density (2.38 t/m³ in this study). Approximately 95% of the asphalt mix mass consists of aggregate, while 5% is asphalt binder. Based on these proportions, the required masses of aggregate, including coarse and fine particles, and asphalt binder were calculated. The energy requirements were calculated for the production of the required construction materials, including the asphalt binder and aggregate materials through values provided in Table 4. The requirement energy for heating up the aggregate and the asphalt binder to mixing temperature was calculated using Equations 9 through 12, developed by Jamshidi 2013; Waples and Waples (2004):

$$Q = \sum_t^T m_i C \Delta \theta \quad (9)$$

where Q is total heat energy requirement, m_i is mass of materials, C is the specific heat capacity of materials (J/kg°C) Δ is the difference between the ambient temperature (25°C) and the mixing temperature (°C)

$$C_{pnt} = 8.95 \times 10^{-3} T^3 - 2.13 \times 10^{-6} T^2 + 0.00172 T + 0.176 \quad (10)$$

$$C_{pt2} = C_{PT} \times \frac{C_{Pnt2}}{C_{Pnt1}} \quad (11)$$

$$C^* = \int_t^T C_{pnt} dt \quad (12)$$

Where C_{Pnt} is the unitless specific heat capacity of aggregate, C_{pnt1} is the unitless specific heat capacity of aggregate at the ambient temperature; C_{pnt2} is the unitless specific heat capacity of aggregate at the mixing temperature, C_{pt1} and C_{pt2} are specific heat

capacity of aggregate at the ambient and mixing temperature (J/kg°C)

The mixing and compaction temperatures of the asphalt mixtures were determined based on the Asphalt Institute's recommendation ("Superpave Performance Graded Asphalt Binder Specifications and Testing. Asphalt Institute" 2001) (SP – 1, 2001)

- Mixing temperature: the temperature corresponds to a viscosity range of 170 ± 20 mPa.s
- Compaction temperature: the temperature corresponds to a viscosity range of 280 ± 30 mPa.s.

To estimate the construction temperature, the recovered binder from different RAP sources was blended with the fresh binder at 140°C in a proportion of 10% to 40% at increments of 10% by mass of binder. Then, the viscosity of the binder blends was tested using a Brookfield rotational viscometer between 120°C to 200°C to meet the viscosity recommended by the Asphalt Institute.

After calculating the total heat energy requirement for asphalt mixes containing RAP across various stages of pavement construction – including aggregate and binder production, aggregate drying, heating of aggregate and asphalt binder to the mixing temperature (as per Equation 9), mix production, and asphalt placement – the fuel requirements for different fuel types and the associated carbon footprint were determined using the proximity factors provided in Table 5 based on Defra (2008).

Figure 5 presents the schematic flowchart of this research. The study begins with laboratory work involving the preparation of asphalt mix samples containing varying RAP contents and sources, which were tested using the SPT, and was explained previously. The resulting data were screened, processed, and used to train ML models for predictive analysis. Subsequently, the required pavement thickness was designed based on the E^* obtained from SPT results. For the sustainability analysis, the mass of required materials was calculated, followed by an estimation of the heat energy demand. Finally, sustainability was evaluated through an analysis of different types of industrial fuel consumption and carbon footprint.

3. Results and discussion

3.1. Analysis of models

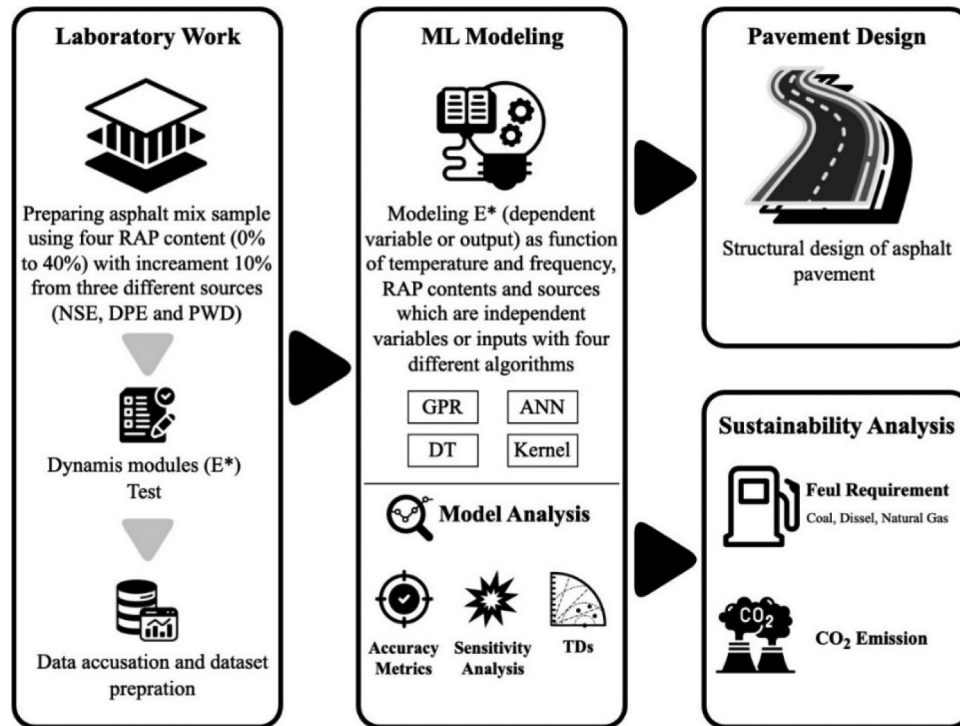
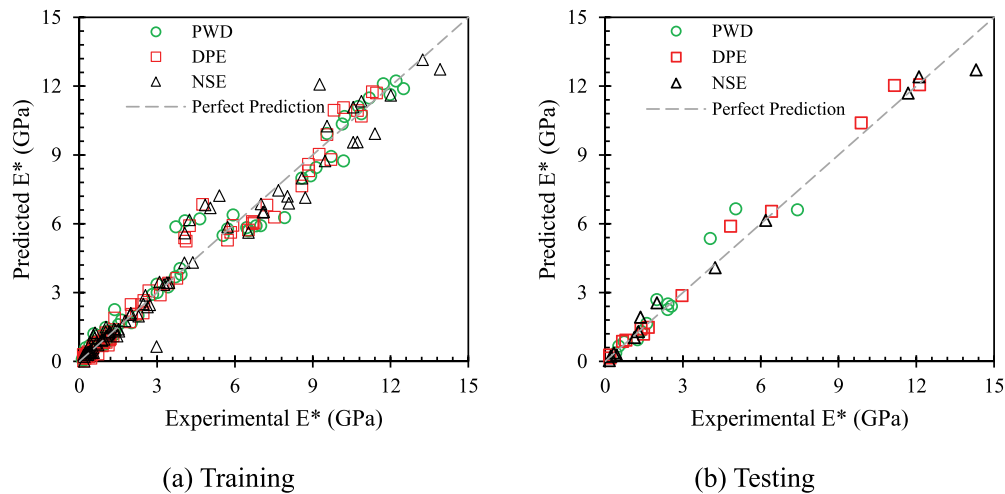
Figures 6–9 present the final outputs of the training and testing process for different RAP sources. The correlation diagrams show a comparison between the outputs of the models based on the adopted ML models. From Figures 6(a), 7(a), 8(a) and 9(a), the data of the model trained using the ANN shows the lowest fluctuation around the perfect line compared with the

Table 4. Energy requirement for asphalt pavement construction.

Material	Energy requirement	Reference
Aggregate production	53 MJ/ton	(NCSA 1977)
Asphalt binder production	6905 MJ/ton	(Stammer and Stodolsky 1995)
Drying aggregate and mix production	350 MJ/ton	(Ang, Fwa, and Ng 1993)
Asphalt placement	13.4 MJ/ton	(Zapata and Gambatese 2005)

Table 5. Conversion factors of the industrial fuel types and carbon footprint.

Fuel type	Proximity factor from heat energy to fuel	Proximity factor from the fuel to carbon footprint
Coal	0.00004263 ton/MJ	3164.3 kgCO ₂ /unit
Diesel	0.00002181 ton/MJ	2295.3 kgCO ₂ /unit
Natural gas	0.02610824 m ³ /MJ	2.023 kgCO ₂ /unit


Figure 5. Schematic illustration of research processes.

Figure 6. Correlation between experimental and predicted E^* with GPR model.

other models, meaning that the ANN model shows more stability than the other ML models tested. The same trend can be seen in Figure 7(b), which implies higher precision in the ANN model. To confirm the trend, the details of the accuracy metrics of all the models and RAP sources in the testing and training procedures are presented in Table 6. From the table, it can be seen that the lowest RMSE in the training

process belongs to the ANN methodology. However, the values of RMSE depend on the RAP source. In the ANN, the lowest RMSE belongs to the PWD (0.27), followed by the DPE (0.29), both highlighted in yellow, while the highest RMSE belongs to the NSE, shown in orange. In the training process, the lowest RMSE after the ANN belongs to the Tree model, followed by the Kernel and GPR models, respectively,

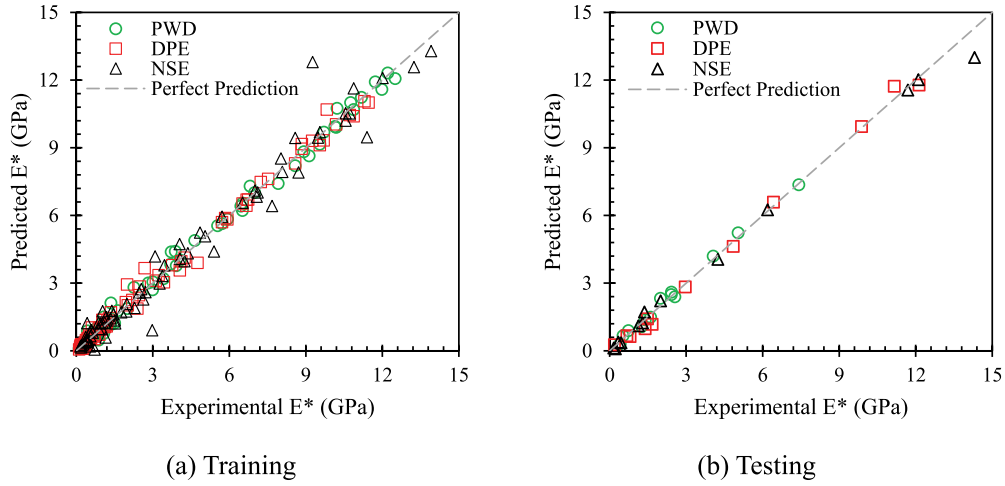


Figure 7. Correlation between experimental and predicted E^* with ANN model.

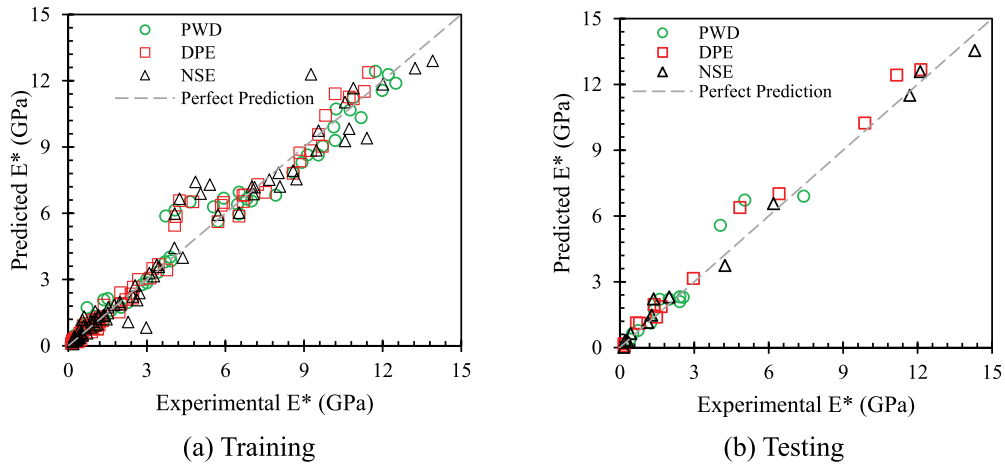


Figure 8. Correlation between experimental and predicted E^* with Kernel model.

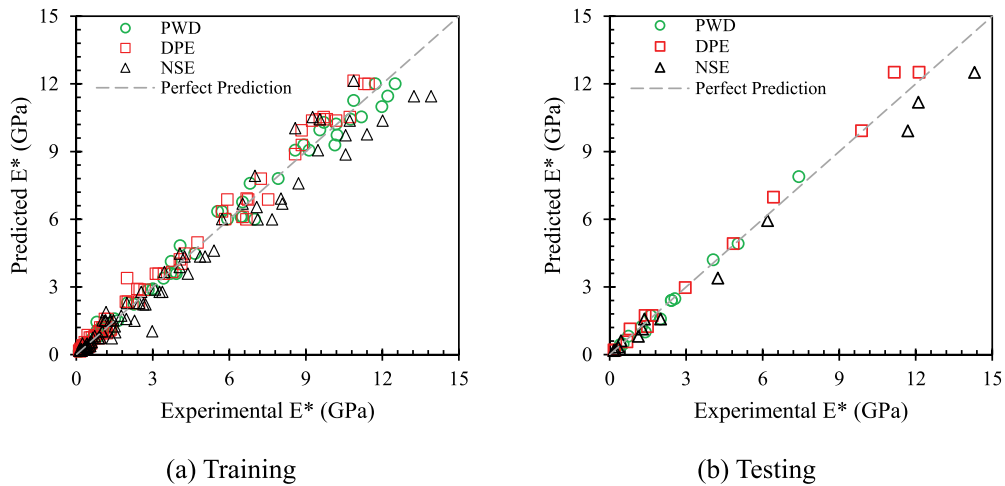


Figure 9. Correlation between experimental and predicted E^* with Tree model.

for all the RAP sources. Additionally, the lowest RMSE and the highest R^2 were seen in the PWD, followed by the DPE and the NSE, in the training process. In the testing process, the ANN showed the best-fitted model of all the RAP sources, highlighted in purple. The second-best fitted model is the Tree model, with

average RMSE and R^2 values of 0.49 and 0.98, respectively. The corresponding values for the GPR model are 0.57 and 0.95. Furthermore, the Kernel model showed the lowest accuracy, with average RMSE and R^2 values of 0.57 and 0.95, respectively. It should be noted that the difference between the second-best

Table 6. Accuracy metrics obtained from four different ML models.

MLs	Procedure	PWD		DPE		NSE	
		RMSE	R ²	RMSE	R ²	RMSE	R ²
GPR	Training	0.48	0.99	0.50	0.98	0.86	0.95
ANN		0.27	1.00	0.29	0.99	0.65	0.97
Kernel		0.56	0.98	0.53	0.98	0.83	0.95
Tree		0.35	0.99	0.43	0.99	0.76	0.96
GPR	Testing	0.65	0.88	0.42	0.99	0.51	0.99
ANN		0.16	0.99	0.28	1.00	0.39	0.99
Kernel		0.68	0.87	0.64	0.98	0.41	0.99
Tree		0.22	0.99	0.44	0.99	0.80	0.97

fitted model and the last one in terms of RMSE and R² are 8% and 3%, respectively, which is almost insignificant.

3.2. Sensitivity analysis results

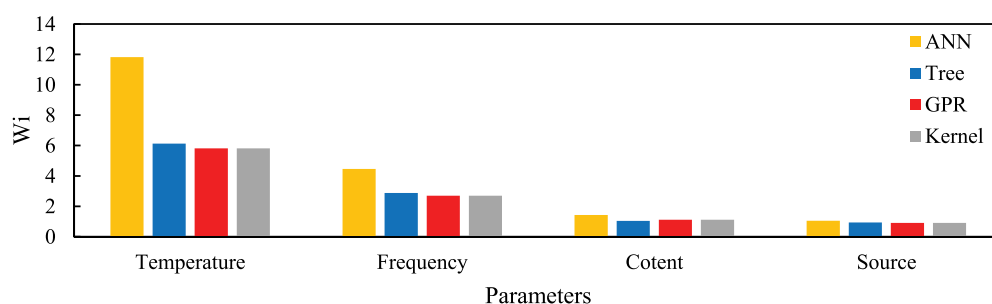
Sensitivity analysis of the models was carried out, and the results are shown in Figure 10. The results of the sensitivity analysis show that temperature had the greatest effect on the results of all models, followed by frequency, RAP content, and source. In other words, temperature was a dominant factor in the outputs of all models, irrespective of RAP source and content. The highest effect of temperature can be seen in the ANN model, while the effect of temperature on the other models is comparable. A similar trend is seen in frequency and RAP content – that is, the ANN model is more sensitive to the frequency. The reason for this phenomenon may be the higher accuracy of the ANN model in comparison with other models. This means that the ANN model can provide more accurate results, as shown in Figure 10; hence, any change in the input of the model may result in the trend of output due to changes in the training data.

3.3. Evaluation of selecting the best model

Figure 11 shows the TDs of the RAP sources for various percentages. For the selection of the best model using TDs, lower SD and higher PCC are desirable because this indicates that the models approach the horizontal axis, which is the output of laboratory testing as a benchmark. In other words, any RAP source results

in the adopted models rotating clockwise towards the horizontal axis indicates that the models can provide more precise outputs. From Figure 11, it can be seen that the closest ML model to the horizontal axis was the ANN for all RAP sources, which is consistent with the results of the accuracy metrics shown in Table 6. For the NSE RAP source (Figure 11(c)), the accuracy of the Tree model and the ANN are comparable. However, it seems that the RMSD of the Tree model is somewhat higher than the ANN model. Figure 11 also shows that the accuracy of the GPR and Kernel models are comparable, depending on the RAP source. As an example, Figures 11(a) and 11(b) show that the GPR and Kernel models are almost equal for the PWD RAP source, while Figure 11(b) shows that the SD of the GPR model is slightly less than that of the Kernel model.

Although these models can yield promising outputs that are comparable with one another, their limitations should not be overlooked. For example, in this study, only one asphalt binder type was used as the base, which may limit the generalisability of the findings of the models. It is recommended that future studies incorporate a variety of asphalt binder types to enhance model robustness. Additionally, only four ML approaches were employed, which may further constrain the study's scope. To broaden the applicability of this research, further recommendations are provided in the 'Suggestions for Future Research' section. One of the inherent limitations of data-driven predictive models is their reliance on high-quality laboratory and/or field data, which are often labour-intensive and time-consuming to obtain. Moreover, the accuracy of ML models heavily depends on the quality and consistency of the input data. Poorly curated or inconsistent data can introduce significant bias, thereby reducing model accuracy. Therefore, rigorous data screening and statistical validation are essential steps prior to model development. It is also important to recognise that different ML algorithms follow distinct philosophies in how they train, process and test data, potentially leading to varying outcomes. This does not necessarily indicate deficiencies or inaccuracies in the modelling process; rather, it highlights the importance of selecting the most appropriate ML approach based on the nature of the input (independent) and output (dependent)

**Figure 10.** Variable sensitive (importance) for the best method.

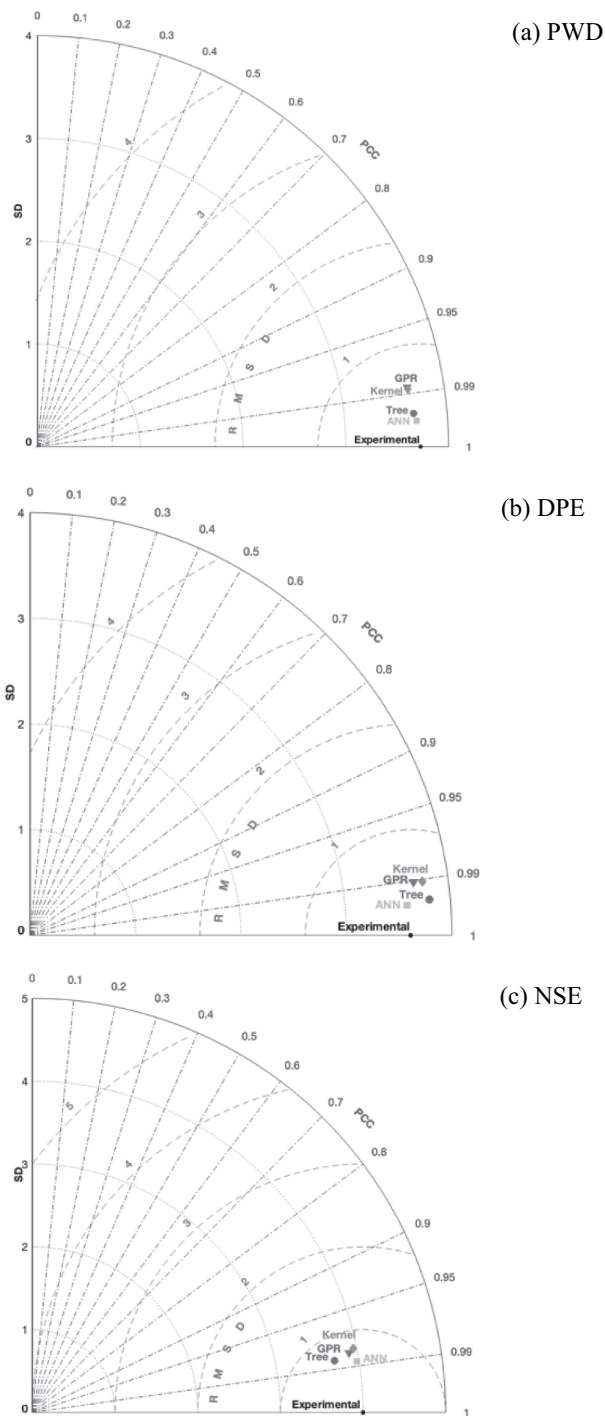


Figure 11. TDs for RAP Mixes in ML Models.

variables, as well as the expertise of the researchers and engineers involved in modelling. Finally, while incorporating a large number of input variables may seem beneficial, it is not always an efficient approach. A complex model with many variables can significantly increase the time required for data processing and model training, potentially limiting its practical application. Furthermore, in many real-world consultancy and research projects, the availability of detailed input data may be limited. Therefore, the complexity and data requirements of ML models should be carefully considered to avoid overfitting and ensure broader usability.

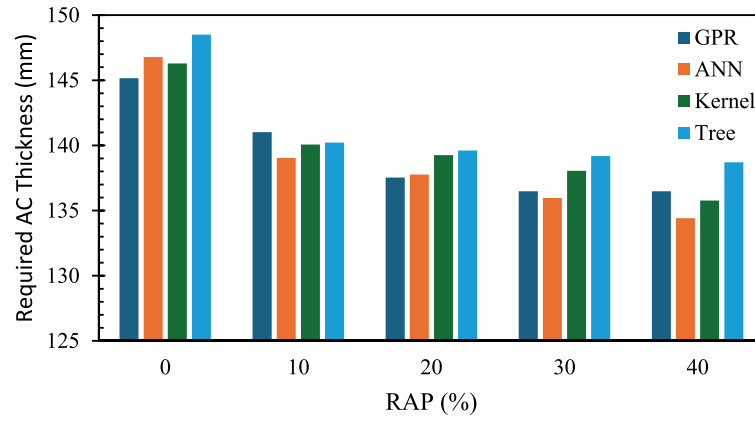
3.4. The effect of RAP source and content on pavement design

The required thickness of the asphalt mix for various sources and RAP content are shown in Table 7. It can be seen that the incorporation of RAP in the asphalt mix may decrease the required asphalt thickness of the surface course because of a higher stiffness in the mix, which results in savings in materials. For example, the required thickness of the asphalt mixture containing 40% RAP is 11 to 12 mm less than the control asphalt mixture, which results in a huge material saving for 1 km of road. From the table, it can be seen that the RAP source has no significant effect on the required thickness of asphalt pavement. For example, the required thickness of asphalt mixture containing 10% RAP from the PWD or the DPE is 141 mm, which is comparable with 143 mm for the asphalt mixture containing RAP from the NSE. Although the required thicknesses of the asphalt mixtures are comparable, the rheological properties of the binder blends and the field performance of the mixtures may be different, which requires further investigation. Additionally, a tiny difference in the engineering properties of the asphalt mixtures can lead to different behaviour in the life span of the asphalt pavement.

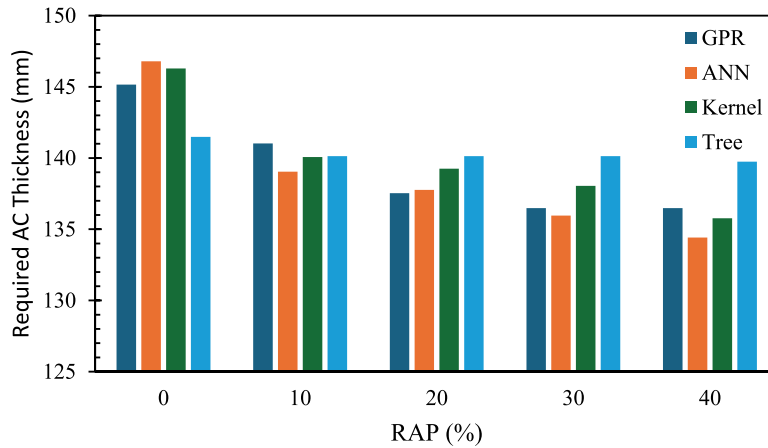
Figure 12 shows the required thickness based on the E^* values outlined by various ML models. From the figure, it can be seen that the required thicknesses yielded from the different ML models are comparable. For example, the required thickness of asphalt pavement containing 20% RAP from the NSE and using the GPR model (Figure 12(a)) is 137.53 mm, and the corresponding values for the DPE and PWD sources are 141.42 mm and 139.66 mm, respectively. Additionally, the required thicknesses outlined by the various ML methods show a decreasing trend as RAP content increases, irrespective of RAP source. In other words, the higher E^* due to the incorporation of RAP, which results in a lower asphalt pavement thickness, can be well understood by the algorithms of the ML methods. It should be noted that such a comparable trend observed in this ML model development might not be universal. Therefore, further investigation is required.

Table 7. Required asphalt thickness of mixtures containing different RAP contents from various sources.

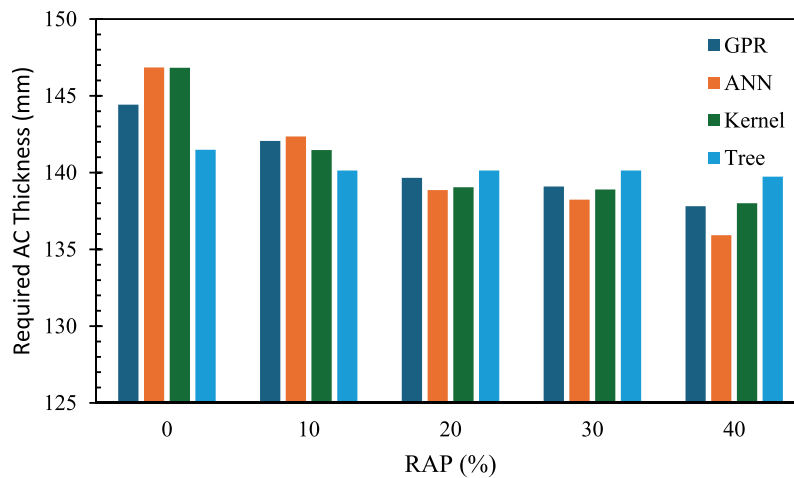
RAP Source	RAP content (%)	Required Thickness (mm)
PWD	0	148
	10	141
	20	138
	30	137
	40	136
DPE	10	141
	20	138
	30	137
	40	136
NSE	10	143
	20	138
	30	137
	40	137



(a) NSE



(b) DPE



(c) PWD

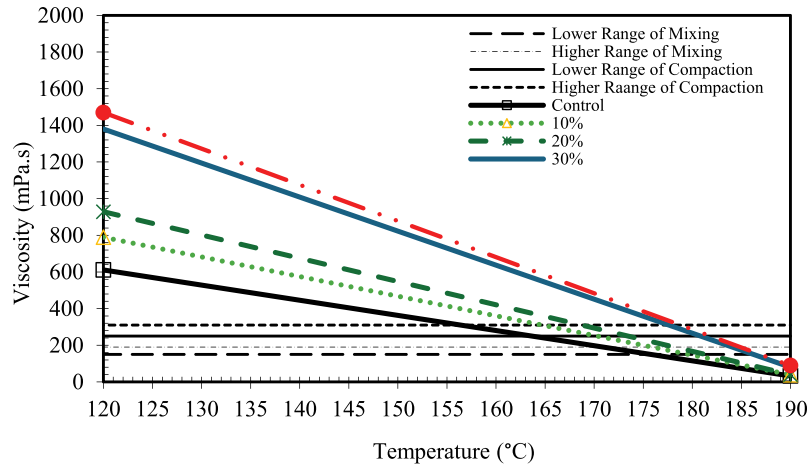
Figure 12. Required Thickness of Asphalt Pavement Layer for the Mixtures Containing RAP from Different Sources.

3.5. The effect of RAP source and content on sustainability

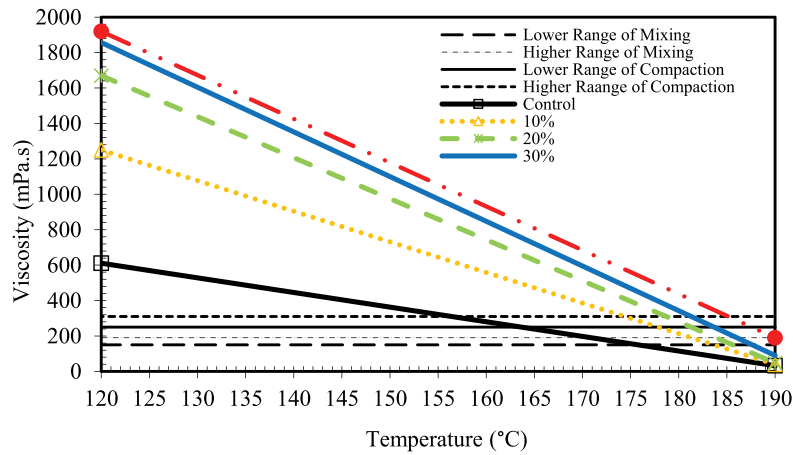
Figure 13 shows the range of mixing and compaction temperatures based on the range of viscosity recommended by the Asphalt Institute. The construction temperatures are shown in Table 8.

Table 8 shows the construction temperatures of the control (0% RAP) and RAP mixtures. The incorporation of RAP increases the mixing and compaction

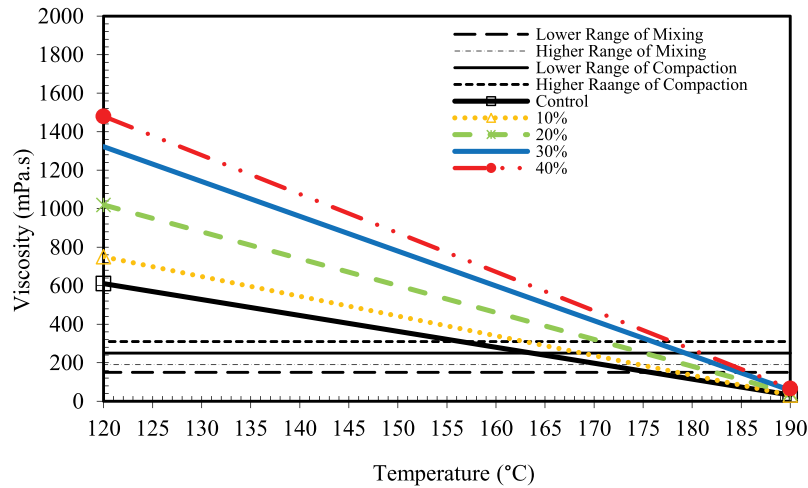
temperatures of the mixes, irrespective of RAP content and source. The reason for this is that the binder in the RAP, which has experienced long-term ageing, increases the viscosity of the resultant asphalt binder blends; therefore, the higher the RAP content, the higher the mixing and compaction temperatures. The minimum increase in the mixing temperature is in the mixes containing 10% RAP, which is 11°C to 17°C higher than the mixing temperature of the control mix. The maximum mixing



(a) NSE



(b) DPE



(c) PWD

Figure 13. Determination of construction temperatures.

temperatures can be seen in the mixes containing 40% RAP, at 20°C to 25°C higher than the control mix. However, the increase in the construction temperature is not equal in the mixes. Additionally, Table 8 shows that the maximum construction temperatures belong to mixes containing DPE RAP. For example, the mixing temperature of the mix containing 10% RAP, highlighted in yellow in the table, is 17°C higher than that of the

control mix (165°C). In other words, the mixing temperature of the mix containing 10% DPE is 10% higher than the control mixture. It can be interpreted that the incorporation of 1% RAP from the DPE results in a 1% increase in the mixing temperature of the mix.

The increase in mixing and compaction temperatures also depends on the RAP source. For instance, 10% RAP from the NSE can increase the mixing tem-

Table 8. Construction temperatures of the control and RAP mixtures.

RAP content (%)	RAP Source	Construction Temperature (°C)	
		Mixing	Compaction
0 (control)	-	165	155
10	NSE	176	165
20		179	170
30		183	175
40		185	180
10	DPE	182	176
20		185	180
30		187	183
40		190	186
10	PWD	175	165
20		180	170
30		184	175
40		185	178

perature by 7%, while the temperature increase is just 3% when using 10% RAP from the DPE. A similar trend can be seen in the compaction temperature. It should be noted that the higher construction temperatures due to the incorporation of RAP may result in higher fuel consumption in the mixing process, resulting in higher environmental burdens in the asphalt mixing plants. To decrease the construction temperatures of the mixes containing RAP or other waste material, the use of warm mix asphalt (WMA) technology can be considered as a practical option. Such technologies decrease binder viscosity and/or workability of asphalt mixes through different mechanisms. However, WMA technology may incur further costs and environmental emissions because of adding new facilities to the mixing plants and the production of new materials. As a result, the environmental effects and costs of WMA technology require more attention to be given to the lifecycle analysis of asphalt mixtures.

To characterise the effect of RAP content and RAP source in the asphalt mix, the environmental burdens of the mixes in terms of CO₂ emissions for various industrial carbon-based energy carriers used in mixing plants are shown in Table 9. It can be seen that the fuel requirements and CO₂ emissions decrease as RAP content increases. However, it can be seen that the fuel requirements and CO₂ emissions are different for various sources. For example, the amount of coal consumed by 10% RAP from the NSE was 13,135 tons, highlighted in orange, while consumption when using 10% RAP from the DPE (highlighted in green) and the PWD (highlighted in blue) was 12,941 tons. Consequently, the CO₂ emissions of the NSE, the LDP/PWD were 30,149 tons, and 29,704 tons, respectively. Therefore, the incorporation of the same percentage of RAP from different sources changes the construction temperatures, which results in changes in fuel requirements and environmental burdens. Thus, the level of sustainability of asphalt mixes containing RAP may change due to the properties of the RAP. It should be noted that although the incorporation of higher RAP increases construction temperatures due to the stiffened binder in RAP (Table 8), which

results in higher energy consumption and CO₂ emissions, it reduces the energy requirements and environmental burdens of pavement manufacture during the aggregate production phase. This means that the increased emissions due to the higher construction temperatures are offset by the lower energy requirements and CO₂ emissions of less aggregate requirement. For example, the incorporation of 40% RAP increases the construction temperature by 6%, while it decreases the aggregate requirement by 40%, resulting in an approximate energy saving of 36%. Similar trends can be seen in other sources.

The results of the ML models to predict E* were used for the pavement design. The thicknesses of asphalt pavement determined based on the E* values outlined by the ML models were used to determine the energy requirements and CO₂ emissions. The results of the energy requirements and emissions analysis were comparable with the experimental data for various RAP contents and sources. For example, the natural gas requirements and CO₂ emissions of 30% RAP from the NSE using experimental data were 6,301,170 m³ and 12,747 tons, respectively. The corresponding values of the natural gas requirements using the GPR, ANN, Kernel, and Tree methods were 62,94,252 m³, 6,270,271 m³, 6,366,658 m³, and 6,418,772 m³, respectively, highlighted in purple in Table 9. The corresponding CO₂ emissions for the GPR, ANN, Kernel, and Tree methods were 12,733 tons, 12,685 tons, 12,880 tons, and 12,985 ktons, respectively, which are comparable with the results for the experimental data (12,747 tons). The values for the fuel requirements and CO₂ emissions of the GPR, ANN, Kernel, and Tree methods range from -2.66% to +2.16% (NSE), -4.5% to +3.56% (PWD), and -1.38% to +2.64% (PWD). Therefore, the general trend shows that sustainability results in terms of fuel requirements and CO₂ emissions are comparable using the various ML methods. Even though the incorporation of RAP decreases the energy requirements and greenhouse gas emissions, burning the cleaner industrial fuel can lead to more

Table 9. Energy requirement and CO₂ emission of asphalt mixtures.

RA P	Source	ML	Energy (TJ)	Fuel Requirement			CO ₂ Emission (kTon)			
				Coal (Ton)	Diesel (m3)	Natural Gas (m3)	Coal	Diesel	Natural Gas	
0	NSE	Experi- men- tal	327	13956	7140	8547072	32033	22593	17291	
10			308	13135	6720	8044415	30149	21264	16274	
20			269	11493	5880	7039008	26381	18607	14240	
30			241	10289	5264	6301170	23616	16656	12747	
40			209	8911	4559	5457502	20454	14426	11041	
0	DPE		326	13933	7128	8533259	31981	22556	17263	
10			303	12941	6621	7925584	29704	20950	16033	
20			269	11509	5888	7048735	26417	18632	14260	
30			241	10301	5270	6308549	23643	16676	12762	
40			209	8920	4564	5463120	20475	14441	11052	
0	PWD		326	13933	7128	8533259	31981	22556	17263	
10			303	12941	6621	7925584	29704	20950	16033	
20			269	11509	5888	7048735	26417	18632	14260	
30			241	10301	5270	6308549	23643	16676	12762	
40			209	8920	4564	5463120	20475	14441	11052	
0	NSE	GPR	319	13641	6979	8354272	31310	22083	16901	
10			304	12968	6634	7941916	29765	20993	16066	
20			269	11496	5881	7040544	26387	18611	14243	
30			241	10277	5258	6294252	23590	16638	12733	
40			209	8943	4575	5476764	20526	14477	11079	
0	DPE		320	13680	6999	8377868	31399	22146	16948	
10			310	13222	6764	8097353	30347	21404	16381	
20			277	11821	6048	7239684	27133	19137	14646	
30			249	10625	5436	6506859	24386	17200	13163	
40			215	9169	4691	5615609	21046	14844	11360	
0	PWD		318	13571	6943	8311683	31151	21971	16815	
10			306	13063	6683	8000487	29984	21148	16185	
20			273	11674	5973	7149585	26795	18899	14464	
30			245	10474	5359	6414622	24041	16956	12977	
40			211	9030	4620	5530135	20726	14618	11187	
0	NSE	ANN	323	13794	7057	8448082	31662	22331	17090	
10			299	12786	6541	7830407	29347	20699	15841	
20			270	11515	5891	7052319	26431	18642	14267	
30			240	10238	5238	6270271	23500	16575	12685	
40			206	8808	4506	5394099	20216	14259	10912	
0	DPE		316	13509	6911	8273123	31006	21869	16737	
10			309	13183	6745	8073700	30259	21342	16333	
20			275	11733	6003	7185932	26931	18995	14537	
30			248	10574	5410	6475959	24271	17118	13101	
40			212	9042	4626	5537759	20754	14638	11203	
0	PWD		323	13800	7060	8451535	31675	22340	17097	
10			307	13090	6697	8016819	30045	21191	16218	
20			272	11607	5938	7108631	26642	18791	14381	
30			244	10410	5326	6375421	23894	16853	12897	
40			208	8906	4556	5454292	20442	14418	11034	
0	NSE	Kerne- l	322	13747	7033	8419306	31554	22255	17032	
10			302	12880	6590	7888415	29564	20852	15958	
20	DPE	Tree	273	11640	5955	7128596	26717	18843	14421	
30			243	10396	5319	6366658	23861	16829	12880	
40			208	8896	4551	5448272	20419	14402	11022	
0			321	13719	7019	8402040	31489	22210	16997	
10			308	13151	6728	8053989	30185	21290	16293	
20			277	11837	6056	7249411	27169	19163	14666	
30			249	10640	5444	6516543	24423	17226	13183	
40			216	9238	4726	5657744	21204	14955	11446	
0			PWD	323	13798	7059	8450384	31670	22337	17095
10				305	13009	6656	7967259	29860	21060	16118
20	272			11623	5946	7118357	26678	18816	14400	
30	245			10460	5351	6405859	24008	16933	12959	
40	212			9042	4626	5537759	20754	14638	11203	
0	NSE		Tree	327	13955	7139	8546496	32031	22591	17290
10				302	12894	6597	7896862	29596	20874	15975
20		273		11670	5970	7147025	26786	18892	14458	
30		245		10481	5362	6418772	24056	16967	12985	
40		213		9088	4650	5565849	20860	14713	11260	
0	DPE	311		13296	6802	8143056	30519	21525	16473	
10		302		12886	6593	7891794	29577	20861	15965	
20		274		11713	5993	7173646	26885	18963	14512	
30		247		10552	5399	6462585	24221	17083	13074	
40		214		9156	4684	5607583	21016	14823	11344	
0	PWD	327		13955	7139	8546496	32031	22591	17290	
10		304		12993	6648	7957685	29824	21035	16098	
20		274		11714	5993	7174157	26887	18964	14513	
30		247		10556	5401	6464891	24229	17089	13078	
40		214		9156	4684	5607583	21016	14823	11344	

sustainable RAP mixtures. As an example, the CO₂ emissions of mixtures containing 20% RAP from the PWD (using experimental data) burning natural gas emits 14,240 ktons of CO₂, while burning diesel and coal emit 18,607 ktons and 26,381 tons of CO₂, respectively. This means that the use of natural gas results in 24% and 46% less CO₂ emissions than diesel and coal, respectively, which is a huge improvement in the sustainability of asphalt mix production. The same trend can be seen in the results of the ML models.

4. Conclusions

The effects of different RAP content (0%–40%) from three different sources were evaluated on the dynamic modulus and sustainability of asphalt mixtures using ML methods. The adopted ML models were GPR, ANN, Kernel, and Tree methods. Mixing and compaction temperatures were determined based on the Asphalt Institute's recommendations. The results of ML models for different RAP contents and sources showed R² values of 87% to 99% for all the methods in testing and training procedures, which are promising. In addition, the sensitivity analysis showed that frequency and temperature play significant roles in the models, followed by RAP content and source. TDs showed the comparable output of the models with the models developed based on experimental data. The mixing results showed that the construction temperature increased by 11°C to 25°C, depending on RAP source and content. The maximum increase was found in the mixes containing DPE RAP, while the mixes containing RAP from the NSE and the PWD showed similar trends.

The pavement design was carried out for a given traffic and soil strength. It was found that the required asphalt thickness decreases as RAP content increases for all the RAP sources and ML models, due to the higher stiffness of the asphalt mixture. In other words, the ML models showed the same trend observed in the experimental data. Additionally, the lowest required thickness was found for DPE RAP mixes (for both experimental and ML data).

The analysis of the thermal energy for the experimental data and the data in the ML models showed that although the DPE mixes required more energy to heat up the aggregate and asphalt binder to the mixing temperature, they required less aggregate material because the higher E* resulted in a lower required asphalt pavement thickness. Furthermore, the incorporation of RAP resulted in further energy savings and decreased greenhouse gas emissions. The results of the sustainability analysis in terms of industrial fuel requirements and CO₂ emissions showed that all the ML models can provide results with less than ±5% errors in comparison with the fuel and CO₂ emissions of the experimental data. However, it should be noted that the data-driven models

developed in this study still have limitations such as high dependency to accurate laboratory data, and different ML approaches with various philosophy in data training. Therefore, it is essential appropriate ML strategies are adopted in the modelling of structural parameters of asphalt mixes.

Although the incorporation of RAP improved the E* values of asphalt mixtures from different sources, it increased the mixing temperature of the asphalt mixtures by up to 6%. However, the lower required thickness of asphalt concrete resulted in lower aggregate material consumption as non-renewable natural resources. Furthermore, the reduction of required aggregate by up to 40% through the use of RAP can result in a significant material saving, irrespective of RAP source. Meanwhile, the use of cleaner industrial fuel in the mixing plant can boost the level of sustainability by up to approximately 50%. It should be noted that the use of the ML predictive models can result in huge material and time savings in sample preparation, which is cumbersome and labour intensive. The ML models developed in this research can be considered as a platform to produce the predictive models of E* for mixes containing different RAP content from various sources, removing the need for material supply and laboratory testing. As a result, the synergistic effects of the use of RAP, ML models, and cleaner industrial fuels in production plants can result in more energy-efficient and sustainable asphalt mixtures for all RAP sources.

5. Suggestion for further research

Further research is recommended to address current and future requirements as follows:

- In this study, only one base binder (control) was used to prepare RAP asphalt blends. It is recommended that future studies utilise various asphalt base binders with different RAP contents sourced from multiple RAP origins to develop ML-based models. This would enable the models to capture the effects of different binder base types, as well as the synergistic effects of binder type and RAP source and content.
- In this research, E* was used as the structural parameter of HMA. Further research is required to develop models based on additional structural parameters, such as fatigue stiffness and ITS, to provide a deep understanding and universal models.
- It is suggested that a wider range of RAP sources and contents be incorporated into sample preparation and structural testing. This would result in a larger dataset, enhancing the reliability and generalisability of ML approaches.
- Various ML strategies can be employed to model E* as a function of RAP source and content. However, different ML methods may yield results

with differing trends that are not necessarily comparable. Therefore, it is recommended to develop a standardised guideline for selecting ML approaches that provide the most realistic and reliable predictions for structural parameters of HMA or WMA.

- It is further recommended to incorporate additional variables into the models, such as asphalt binder modifiers, construction technologies, aggregate types and gradations, filler types, and various waste material types and contents. Including these factors would improve model robustness and practical applicability of data-driven predictive models of HMA and WMA.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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